RAJALAKSHMI ENGINEERING COLLEGE

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D2

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n

The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

#define MAX_N 1000

unsigned long long dp[MAX_N][MAX_N];
int grid[MAX_N][MAX_N];

unsigned long long max(unsigned long long a, unsigned long long b) {
    return (a > b) ? a : b;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    // Read the grid
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &grid[i][j]);
        }
    }

    // Initialize base case
    dp[0][0] = grid[0][0];

    // Initialize first row
    for (int j = 1; j < n; j++) {
        dp[0][j] = dp[0][j - 1] + grid[0][j];
    }

    // Initialize first column
    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + grid[i][0];
    }
```

```
37
38      // Fill the DP table
39      for (int i = 1; i < n; i++) {
40          for (int j = 1; j < n; j++) {
41              dp[i][j] = grid[i][j] + max(dp[i - 1][j], dp[i][j - 1]);
42          }
43      }
44
45      // Output maximum total monetary value
46      printf("%llu\n", dp[n - 1][n - 1]);
47
48      return 0;
49  }
```

Check

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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